BrickPlan: Diagnosing Future Action Cost and Error Recovery in Visual Assembly Planning

Supplementary Material

A. Exact labels and admissible bounds

The archived predecessor audit evaluates every stored nonnegative state–goal label. Let R be occupied blocker closure for required removals and insertions. Without capacity or inspection obligations, the formula $|A \triangle G| + 2|R \cap A \cap G|$ agrees with all 1,802,728 audited labels. This is a finite-census equality, not a general theorem about arbitrary assembly domains or visual perception.

For BrickPlan's lower bound, initially mismatched sites each require an operation. Every occupied target site in the required removal closure must be detached and reattached. Every unfinished job needs one audit. These counts refer to disjoint required operations. Recursively including occupied path blockers can only add necessary removals. Ignoring capacity scheduling and some connectivity effects relaxes the problem, so $h(x) \leq C^*(x)$. The verifier checks this inequality at every oracle node.

For every nonterminal product node, the certificate checks $C^*(x) = \min_{a \in A(x)}[1 + C^*(T(x, a))]$, with terminal cost zero. Physical edges are compared with independent runtime transition enumeration and inverse execution; audit edges are replayed as well. Every retained product state reaches a terminal. The physical graph is reversible within its capacity-constrained component. Audits add a persistent bit without removing a physical action, and all admitted jobs can be exposed. Consequently, legal successors do not become genuine irreversible dead ends. Budget exhaustion and extra cost remain possible.

B. Construction details

The machine-readable source plan distinguishes development and new-source groups before new-source solver discovery. Legacy tasks retain their earlier selection bias. Additional task generation uses deterministic seeds, sampled states and work orders, and solvability admission. Geometric pruning was refined after an initial collision-rejecting pass, before new-source policy results. Both passes and their exact retained source indices are archived. No difficulty-based rejection is used for these new-source tasks.

Ten geometry pairs share registered pieces and original transforms, start, goal, capacity, jobs, and proposed action sequence. Induced fixed contexts are separately qualified. Four context interventions across source groups 31027, 10014, and 10036 yield these ten pairs, all with unequal fixed-part counts. The remaining 37 qualified context candidates yield no retained answer-changing pair under the declared search. The full census is in data/geometry-probe-audit.json. Original/grayscale pairs preserve all geometry and labels.

The 144 interior-order pairs are balanced over conflict bins 2–5, 6–10, and 11+. Legal and reordered plans share exact prefixes up to the first conflict, endpoints, and action multisets. Legal witnesses have no repeated states. The stored audit exposes each conflict position, so readers can run other prefix controls.

Table 1. A prefix-only software control returns the first conflict if it appears within k executed operations, otherwise "all legal." The denominator is all 288 continuation questions.

Prefix length	Correct / assigned	Accuracy
0	144/288	50.0%
1	144/288	50.0%
2	158/288	54.9%
3	172/288	59.7%
5	192/288	66.7%
8	224/288	77.8%
11	248/288	86.1%
16	273/288	94.8%
24	282/288	97.9%
32	288/288	100.0%

Table 2. Measured finite-horizon choice sets for all 653 next-action questions. Any/all refer to optimistic/pessimistic ties; uniform tie correctness is an exact average, not a sampled policy result.

Depth	Choices	All optimal	Any optimal	Uniform tie
1	653	571	653	94.0%
2	653	592	653	95.4%
4	653	590	653	94.7%
8	653	612	653	96.6%

C. Shallow choice and control diagnostics

The depth-one minimizing set contains an optimal action at all 653 selected checkpoints. Therefore the diagnostic bank does not demonstrate that shallow reasoning necessarily excludes the correct answer. Deeper

Table 3. Structural policy completion by operation-length bin. Each nonoracle policy has three tie-order seeds per episode.

Control	D1	D2	D3	D4
Closure greedy	291/291	449/456	242/249	79/135
Capacity dispatch	291/291	450/456	232/249	66/135
Closure A* (100)	291/291	447/456	225/249	45/135
Closure A* (1,000)	291/291	456/456	240/249	92/135
Closure A* (full)	291/291	456/456	249/249	135/135
Hamming A* (full)	291/291	456/456	249/249	135/135

horizons can change tie composition nonmonotonically. The full table is retained without redefining difficulty to match its results.

All 10,556 raw policy task-run records are archived in data/software-control-records.json.gz. They include one oracle seed and three seeds for nine other policies, with actions, outcomes, expansions, timings, and regrets. The release checker recomputes every overall, tier, source, and seed aggregate. The full search cap of 500,000 is a resource ceiling, not actual expansion usage. Oracle loading and cached local timing are not comparable to calibrated online search.

D. Recovery, diagnosis, and censoring

Traverse only edges with $C^*(T(x,a)) = C^*(x) - 1$ from pre-error state b. The reached states are exactly the union of optimal completions: each traversal is an optimal prefix, and every optimal prefix satisfies that equality. Remove the terminal state to obtain the repair target $\mathcal{C}_b^-$. The generator's delayed checkpoints remain outside this set after an error and lag 1, 3, or 5. Independent forward BFS verifies their shortest return distances, which range from 1 to 7. The 943 questions comprise 337, 307, and 299 checkpoints at those lags.

Each positive-excess legal action creates its own spontaneous event; later actions update every active event. A second error never replaces the first. Immediate recovery and lag-specific delayed cohorts are reported separately. Unfinished events are censored; terminal events that never restore a nonterminal corridor state are observed failures. Reaching a goal after a costly detour earns completion credit, but not automatic corridor-repair credit.

The inverse-last-action control uses only the public executed history and candidate list. It earns zero when no inverse is offered. It solves 235/337 lag-one, 162/307 lag-three, and 150/299 lag-five questions. The combined 547/943 accuracy is 58.0%. There are 368 cases with a legal but suboptimal inverse and 28 with no offered inverse. Lag is thus an experimental descriptor, not proof of necessary nonlocal reasoning.

There are 26 all-correct-choice calibration questions across the diagnostic bank. Their correct tie sets are preserved, and the evaluator reports calibration separately from the 1,926 discriminative rows. The 1,952 rows map to 1,819 canonical problems. Assigned repeated variants are averaged before problems and sources; missing responses receive zero before this averaging. The evaluator exposes both row and clustered scores, plus each inverse-control cohort. None of these private labels is included in the policy observation.

An accepted diagnosis concerns legal positive-excess events since the last diagnosis or episode start. The first later diagnosis is attached to each previously undiagnosed event, with its sign and lag. True positive diagnosis checkpoints open replanning-suffix records. Always-negative or selectively requested diagnoses can yield misleading accuracy, so report event detection and query coverage alongside it.

Plan-survival reports retain the first cost loss even after recovery, and include explicit eligible/observed/unknown counts per operation index. They are descriptive task-budget curves, not Kaplan–Meier population estimates. Conditions should use matched assignments and registered source strata. Missing episodes must not disappear from denominators.

D.1. Regret conservation

For a successful trace, accepted attaches A_c and detaches D_c satisfy $A_c - D_c = \Delta$, the target-minus-start occupancy count. There are exactly u accepted previously unfinished audits. Duplicate audits are rejected. If N_{invalid} is the number of charged invalid attempts, then

$$T = 2D_c + \Delta + u + N_{\text{invalid}}, \tag{1}$$

$$C^*(x_0) = 2D^* + \Delta + u, \tag{2}$$

$$T - C^*(x_0) = 2(D_c - D^*) + N_{\text{invalid}}. \tag{3}$$

This audit-inclusive identity reconciles detachment regret and total excess. Rejected actions consume budget; stale HTTP revisions and malformed request envelopes are rejected before mutation and are not action events.

E. Visual stimuli and actual model calls

All executable geometry uses original source poses and LDraw triangle meshes. The 214 distinct part meshes contain 290,338 triangles. ASSET-LICENSES.json preserves source models, authors, URLs, and part-level notices; attribution and licenses remain separate from code. Qualification proxies do not replace the displayed

Table 4. Metric denominators and treatment of missing outcomes.

Metric	Eligible denominator	Missing / termination rule
Completion / optimal completion	All assigned episodes	Unstarted and unfinished are failures
Successful mean regret	Successfully finished episodes	Null for failures; report success coverage
Avoidable-cost choice	Legal decisions with zero- and positive-excess alternatives	Invalid attempts excluded here, charged in total
Budget-loss choice	Viable checkpoints with viable and nonviable legal choices	Other budget losses remain in raw records
True-unreachability choice	No eligible choices in retained components	Undefined, not empirical evidence of zero risk
Recovery bounds	Every eligible legal positive-excess event	Lower bound restored/all; upper adds censored
Diagnosis accuracy	Accepted diagnosis queries	Unqueried intervals are not inferred correct
Joint pair correctness	Fully assigned pairs	Missing member prevents joint credit
Discriminative probe accuracy	Assigned noncalibration questions of that family	Missing or invalid response gets no credit
Source macro accuracy	Variants within problem, then problems within source, then sources	Missing variants enter as zero before averaging
Calibration accuracy	Assigned all-correct-choice questions	Report separately; not evidence of discrimination

shapes. Figure provenance records exact states, seeds, bank hash, and image hashes.

The public renderer supersamples by two before Lanczos downsampling to native HD. Display canvases account for device-pixel ratio, and native zoom avoids CSS-only enlargement of low-resolution pixels. Catalog tiles are 480×360 and catalogs are 1920 pixels wide. Models receive image content, not text describing images. Preprocessing is separately declared: the local SmolVLM case resizes to longest edge 512, with splitting disabled. HD display does not imply native-HD model processing.

The preserved version-2.1 local case contains four actual inference calls on historical public E069, seed 0. The original task and bank hash are archived; this trace is not relabeled as version-2.2 evaluation. All calls return descriptive prose instead of an executable JSON action. The executor rejects each, and the controller stops at its four-call limit. Every prompt, raw response, image hash, latency, measured token count, action, and feedback is archived. No oracle action or fabricated explanation replaces model output. This is a failed integration example. It supplies neither an estimate of SmolVLM accuracy nor evidence about larger VLMs.

F. Protocol and reproducibility

Study runs bind participant, condition, seed, assignments, and bank hash. The evaluator replays mutations, validates views, and normalizes candidate permutations. Retries are idempotent across restart; changed content or stale revisions are rejected. Failed atomic persistence rolls back mutations.

Practice exposes rich catalogs, solution review after submission, and free 3D inspection. Study mode blocks these and private downloads. Portable evaluator data must remain on the evaluator host during blind studies; a local working-directory convention is not a security sandbox.

The practice model manager supports local MLX and OpenAI-compatible vision calls, durable outputs, and bounded run/step/stop. Credentials never enter browser exports. Stop prevents subsequent actions, not in-flight provider billing. Interrupted calls are never silently retried.

The population study runner is distinct. Its reported-usage mode reserves maximum tokens and cost before calls; adapters account for images and attest usage. Missing accounting, timeout, malformed responses, or limit violations stop continuation. Unavailable-accounting mode marks tokens and currency unavailable. This relies on adapter/provider reports, not a provider-enforced spending quota.

The main paper uses the unmodified official CVPR author-kit, pinned to commit 2917585, labeled 2026. A future year's kit is not claimed. The PDF uses public preprint mode without margin/font compression. Review mode requires the actual year and paper ID.

G. Base source assemblies

The table lists 17 base assemblies from 16 kit groups; roadster and kart share kit 31027. An additional 82 qualified context profiles are indexed in the machine-readable source census. They are not independent sources. Profiles without episodes are zero-yield candidates, not benchmark tasks. Registered and movable counts describe different quantities.

Base assembly	Kit	Parts	Registered	Movable	Episodes
Roadster body	31027	37	17	17	14
Kart deck	31027	25	18	18	14
Airframe	31028	24	21	21	14
Truck cab and roof	10156	76	18	18	10
House roof assembly	31009	221	18	17	14
Sports car upper body	5867	149	22	10	3
Source 10001 subassembly	10001	17	2	2	8
Source 10159 subassembly	10159	177	16	16	12
Source 10220 subassembly	10220	153	16	16	12
Source 10014 subassembly	10014	71	16	15	12
Source 10128 subassembly	10128	98	16	14	12
Source omr-31031 subassembly	31031	42	12	12	12
Source 10036 subassembly	10036	79	16	16	48
Source 10213 subassembly	10213	179	2	2	28
Source omr-10242 subassembly	10242	180	16	16	48
Source omr-10265 subassembly	10265	179	4	4	48
Source omr-10269 subassembly	10269	105	16	11	48

Partition	Sources	Episodes	Greedy	Full A*
Development	8	109	87.2%	100.0%
Held-out source	8	268	96.5%	100.0%

Version 2.2 admitted five of eleven additional frozen source candidates. The other six remain in the source-plan rejection census. Only source geometry and solvability determine this admission; no policy-performance threshold is used. All earlier episode and diagnostic definitions are preserved. New annotations specify scoring roles and canonical identities.

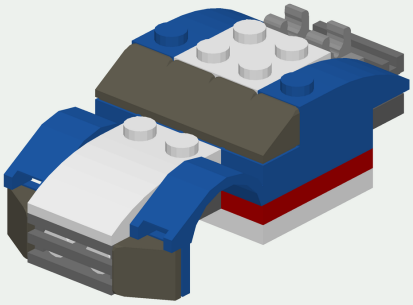
H. Static and active view certificates

Static diagnostics supply four fixed views at revision zero: isometric, left, rear, and right. All five panels are included for each view, and later acquisition is disabled. The structured diagnostic condition uses the same initial views. Active image diagnostics begin with isometric only; each inspect increments the view budget. These are separate conditions rather than interchangeable names for one observation.

The native-resolution certificate compares every geometry pair in all twenty initial panels. All ten pairs have at least one pixel-distinct view; three pairs have identical isometric panels. The certificate is bound to the bank hash and identity seed. It demonstrates input distinguishability, not human agreement or perceptual answerability. Other image-processing pipelines and identity seeds require their own checks. Study results must report preprocessing and acquisition counts.

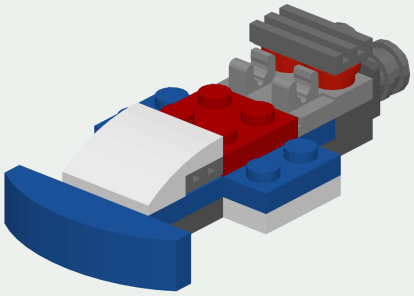
I. Original-source gallery: assemblies 1–6

Kit 31027



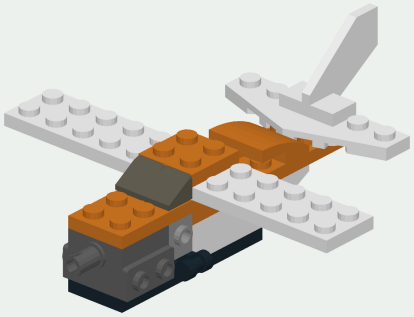
37 source parts; 17 registered
17 movable in the full-assembly component

Kit 31027



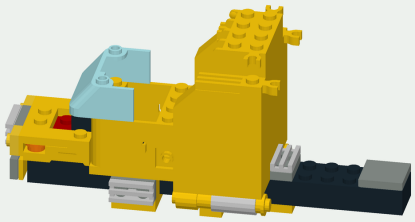
25 source parts; 18 registered
18 movable in the full-assembly component

Kit 31028



24 source parts; 21 registered
21 movable in the full-assembly component

Kit 10156



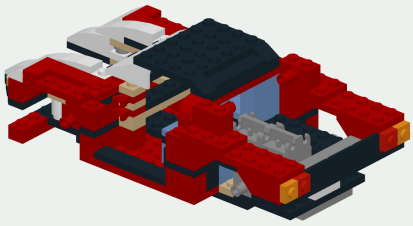
76 source parts; 18 registered
18 movable in the full-assembly component

Kit 31009



221 source parts; 18 registered
17 movable in the full-assembly component

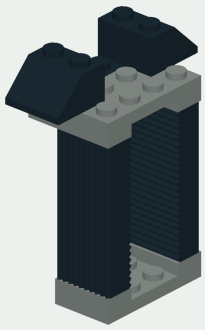
Kit 5867



149 source parts; 22 registered
10 movable in the full-assembly component

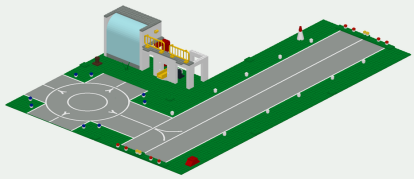
J. Original-source gallery: assemblies 7–12

Kit 10001



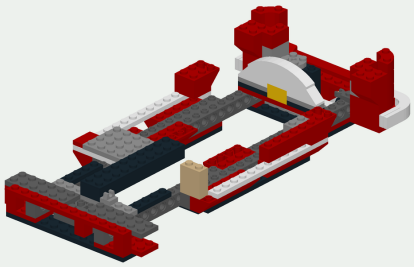
17 source parts; 2 registered
2 movable in the full-assembly component

Kit 10159



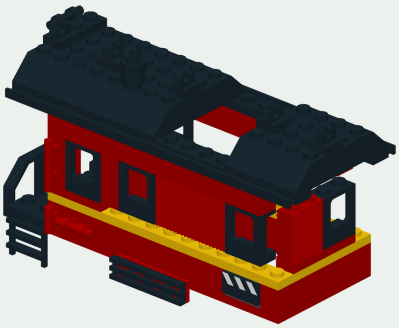
177 source parts; 16 registered
16 movable in the full-assembly component

Kit 10220



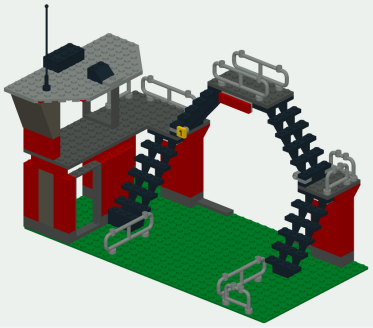
153 source parts; 16 registered
16 movable in the full-assembly component

Kit 10014



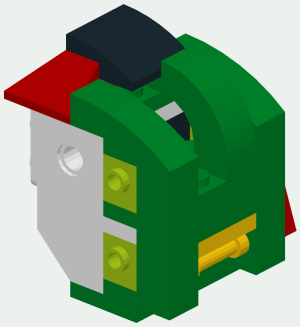
71 source parts; 16 registered
15 movable in the full-assembly component

Kit 10128



98 source parts; 16 registered
14 movable in the full-assembly component

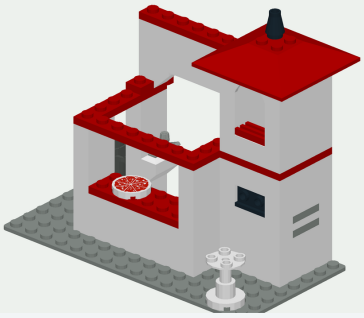
Kit 31031



42 source parts; 12 registered
12 movable in the full-assembly component

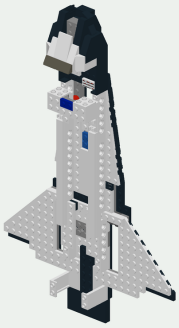
K. Original-source gallery: assemblies 13–17

Kit 10036



79 source parts; 16 registered
16 movable in the full-assembly component

Kit 10213



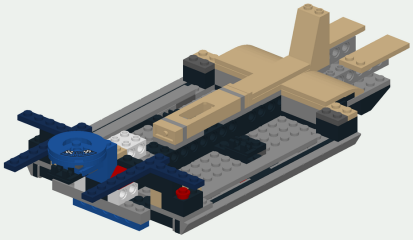
179 source parts; 2 registered
2 movable in the full-assembly component

Kit 10242



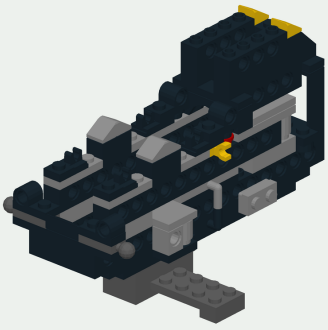
180 source parts; 16 registered
16 movable in the full-assembly component

Kit 10265



179 source parts; 4 registered
4 movable in the full-assembly component

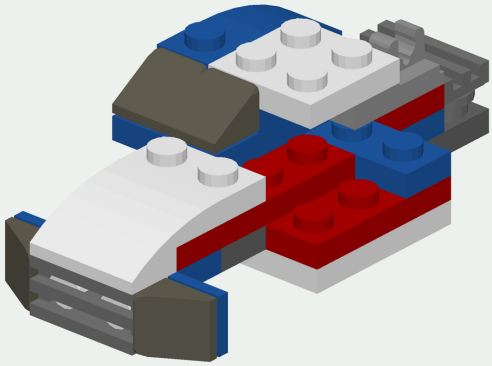
Kit 10269



105 source parts; 16 registered
11 movable in the full-assembly component

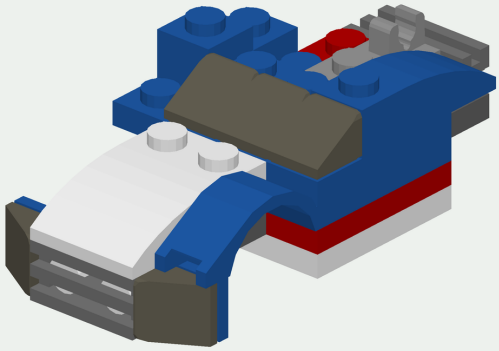
L. A certified optimal resource replay

Operation 00



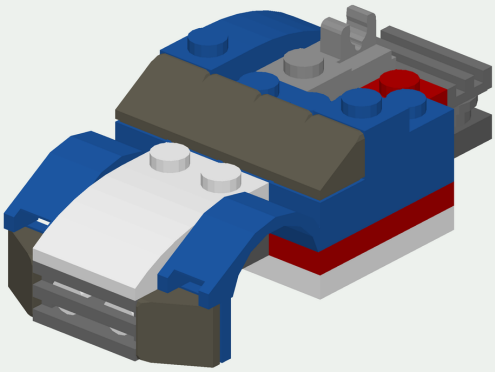
Rack 9/9; remaining cost 42
8 registered pieces installed

Operation 14



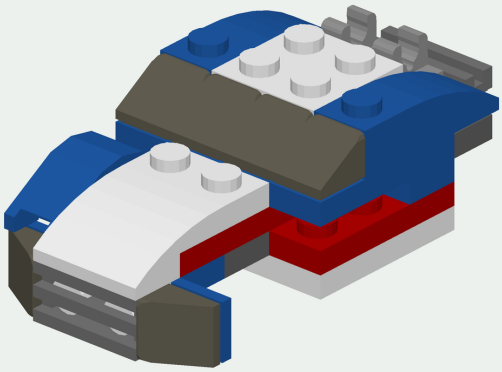
Rack 5/9; remaining cost 28
12 registered pieces installed

Operation 28



Rack 3/9; remaining cost 14
14 registered pieces installed

Operation 42



Rack 1/9; remaining cost 0
16 registered pieces installed

This is an exact software-oracle witness, not a VLM trajectory. All displayed states preserve source poses and certified connectivity. The figure provenance contains the task ID, seed, occupancy, ledger, step indices, and exact remaining cost. Rack holders and off-assembly work are explicit abstractions of the task.

M. Complete episode census

"Jobs" is the declared inspection count, "Done" its initially completed subset, and A* expansions use seed 0 with the closure bound. D1–D4 are length bins. Internal IDs are evaluator documentation; participant IDs are consistently permuted by seed. Exact starts, goals, jobs, witnesses, and oracle paths are available in the task bank.

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
roadster-body-01	D3	17	0	0	17	17	17
roadster-body-02	D4	17	0	0	32	32	32
roadster-body-03	D4	9	0	0	27	23	33
roadster-body-04	D4	9	0	0	27	23	38
roadster-body-05	D4	9	0	0	42	24	2555
roadster-body-06	D4	9	0	0	42	24	2555
roadster-body-07	D4	9	2	0	37	27	2185
roadster-body-08	D4	9	2	0	37	27	2139
roadster-body-09	D3	9	2	1	20	20	21
roadster-body-10	D3	9	2	1	18	18	18
roadster-body-11	D3	9	1	0	18	18	18
roadster-body-12	D3	9	1	0	18	18	18
roadster-body-13	D4	9	4	0	39	29	2131
roadster-body-14	D4	9	4	0	39	29	2129
kart-deck-01	D3	18	0	0	18	18	18
kart-deck-02	D4	18	0	0	25	25	25
kart-deck-03	D3	5	0	0	17	11	32
kart-deck-04	D3	5	0	0	17	11	32
kart-deck-05	D4	5	0	0	26	18	119
kart-deck-06	D3	5	0	0	24	16	90
kart-deck-07	D4	5	2	0	31	21	121
kart-deck-08	D4	5	2	0	29	21	218
kart-deck-09	D3	5	2	1	21	15	68
kart-deck-10	D3	5	2	1	23	19	42
kart-deck-11	D3	5	1	0	23	19	42
kart-deck-12	D3	5	1	0	21	15	68
kart-deck-13	D4	5	4	0	33	23	403
kart-deck-14	D4	5	4	0	33	23	392
airframe-01	D3	21	0	0	21	21	21
airframe-02	D4	21	0	0	33	33	33
airframe-03	D2	8	0	0	16	14	16
airframe-04	D2	8	0	0	16	14	16
airframe-05	D3	8	0	0	17	13	30
airframe-06	D3	8	0	0	17	13	30
airframe-07	D3	8	2	0	22	20	22
airframe-08	D3	8	2	0	22	20	22
airframe-09	D2	8	2	1	9	9	9
airframe-10	D2	8	2	1	9	9	9
airframe-11	D2	8	1	0	9	9	9
airframe-12	D2	8	1	0	9	9	9
airframe-13	D4	8	4	0	26	24	26
airframe-14	D4	8	4	0	26	24	26
truck-cab-01	D3	18	0	0	18	18	18
truck-cab-02	D4	18	0	0	27	27	27
truck-cab-03	D4	12	2	0	40	34	1522
truck-cab-04	D4	12	2	0	36	30	809
truck-cab-05	D3	12	2	1	21	19	22
truck-cab-06	D3	12	2	1	21	19	22
truck-cab-07	D3	12	1	0	21	19	22
truck-cab-08	D3	12	1	0	21	19	22
truck-cab-09	D4	12	4	0	42	36	1546

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
truck-cab-10	D4	12	4	0	38	32	831
house-roof-01	D3	18	0	0	18	18	18
house-roof-02	D3	18	0	0	22	22	22
house-roof-03	D3	5	0	0	21	17	37
house-roof-04	D3	5	0	0	21	17	37
house-roof-05	D3	5	0	0	22	18	45
house-roof-06	D3	5	0	0	22	18	45
house-roof-07	D4	5	2	0	27	23	54
house-roof-08	D4	5	2	0	27	23	50
house-roof-09	D3	5	2	1	17	17	19
house-roof-10	D3	5	2	1	17	17	23
house-roof-11	D3	5	1	0	17	17	23
house-roof-12	D3	5	1	0	17	17	19
house-roof-13	D4	5	4	0	33	29	76
house-roof-14	D4	5	4	0	31	27	68
sports-body-01	D2	22	0	0	10	10	10
sports-body-02	D2	22	0	0	11	11	11
sports-body-03	D3	8	2	0	17	17	17
source-10001-01	D1	2	0	0	2	2	2
source-10001-02	D1	2	0	0	1	1	1
source-10001-03	D1	2	2	0	5	5	5
source-10001-04	D1	2	2	0	6	6	6
source-10001-05	D1	2	0	0	2	2	2
source-10001-06	D1	2	0	0	1	1	1
source-10001-07	D1	2	2	0	4	4	4
source-10001-08	D1	2	2	0	4	4	4
source-10159-01	D2	10	0	0	11	11	11
source-10159-02	D2	10	0	0	9	9	9
source-10159-03	D2	10	2	0	12	12	12
source-10159-04	D2	10	2	0	11	11	11
source-10159-05	D3	10	4	0	21	21	21
source-10159-06	D2	10	4	0	14	14	14
source-10159-07	D2	10	0	0	10	10	10
source-10159-08	D2	10	0	0	14	12	45
source-10159-09	D2	10	2	0	14	14	14
source-10159-10	D2	10	2	0	9	9	9
source-10159-11	D2	10	4	0	12	12	12
source-10159-12	D2	10	4	0	16	16	16
source-10220-01	D1	10	0	0	8	8	8
source-10220-02	D1	10	0	0	7	7	7
source-10220-03	D2	10	2	0	14	14	14
source-10220-04	D2	10	2	0	13	13	13
source-10220-05	D2	10	4	0	14	14	14
source-10220-06	D3	10	4	0	19	19	19
source-10220-07	D1	10	0	0	7	7	7
source-10220-08	D2	10	0	0	11	11	11
source-10220-09	D2	10	2	0	12	12	12
source-10220-10	D2	10	2	0	10	10	10
source-10220-11	D2	10	4	0	14	14	14
source-10220-12	D2	10	4	0	14	14	14
source-10014-01	D2	10	0	0	10	10	10
source-10014-02	D3	10	0	0	21	21	22
source-10014-03	D2	10	2	0	10	10	10
source-10014-04	D3	10	2	0	20	20	20
source-10014-05	D4	10	4	0	26	26	29
source-10014-06	D3	10	4	0	21	21	21

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-10014-07	D1	10	0	0	8	8	8
source-10014-08	D2	10	0	0	15	15	15
source-10014-09	D2	10	2	0	14	12	58
source-10014-10	D2	10	2	0	14	14	15
source-10014-11	D3	10	4	0	21	17	4372
source-10014-12	D2	10	4	0	15	13	52
source-10128-01	D1	10	0	0	5	5	5
source-10128-02	D2	10	0	0	14	14	14
source-10128-03	D2	10	2	0	15	15	15
source-10128-04	D3	10	2	0	18	18	18
source-10128-05	D3	10	4	0	24	24	24
source-10128-06	D3	10	4	0	18	18	18
source-10128-07	D2	10	0	0	10	10	10
source-10128-08	D1	10	0	0	8	8	8
source-10128-09	D2	10	2	0	11	11	11
source-10128-10	D2	10	2	0	14	14	22
source-10128-11	D2	10	4	0	15	15	15
source-10128-12	D3	10	4	0	19	19	19
source-omr-31031-01	D2	8	0	0	12	12	12
source-omr-31031-02	D1	8	0	0	6	6	6
source-omr-31031-03	D2	8	2	0	13	13	13
source-omr-31031-04	D2	8	2	0	11	11	11
source-omr-31031-05	D3	8	4	0	17	17	17
source-omr-31031-06	D3	8	4	0	20	20	20
source-omr-31031-07	D2	8	0	0	13	11	26
source-omr-31031-08	D2	8	0	0	15	15	23
source-omr-31031-09	D1	8	2	0	8	8	8
source-omr-31031-10	D2	8	2	0	10	10	10
source-omr-31031-11	D2	8	4	0	14	14	14
source-omr-31031-12	D2	8	4	0	15	15	15
roadster-body-context-0-01	D4	9	1	0	34	32	954
roadster-body-context-1-01	D4	9	1	0	34	32	953
roadster-body-context-0-02	D4	9	1	0	34	30	1018
roadster-body-context-1-02	D4	9	1	0	34	30	1017
roadster-body-context-0-03	D4	9	1	0	34	30	847
roadster-body-context-1-03	D4	9	1	0	34	30	846
roadster-body-context-0-04	D4	9	1	0	32	30	414
roadster-body-context-1-04	D4	9	1	0	32	30	413
source-10036-01	D2	10	0	0	16	16	16
source-10036-02	D1	10	0	0	8	8	8
source-10036-03	D2	10	0	0	12	12	12
source-10036-04	D4	10	0	0	29	23	406
source-10036-05	D2	10	2	0	16	16	16
source-10036-06	D3	10	2	0	17	17	17
source-10036-07	D3	10	2	0	17	17	17
source-10036-08	D2	10	2	0	15	15	15
source-10036-09	D4	10	4	0	27	27	27
source-10036-10	D3	10	4	0	24	24	24
source-10036-11	D4	10	4	0	26	26	26
source-10036-12	D4	10	4	0	34	28	7595
source-10036-13	D2	10	0	0	11	11	11
source-10036-14	D2	10	0	0	10	10	10
source-10036-15	D2	10	0	0	15	15	27
source-10036-16	D2	10	0	0	12	12	12
source-10036-17	D2	10	2	0	16	16	22
source-10036-18	D4	10	2	0	26	16	3168

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-10036-19	D3	10	2	0	24	14	2226
source-10036-20	D2	10	2	0	14	14	21
source-10036-21	D4	10	4	0	30	20	12578
source-10036-22	D3	10	4	0	24	18	1714
source-10036-23	D3	10	4	0	19	19	27
source-10036-24	D3	10	4	0	20	20	56
source-10036-25	D2	10	0	0	10	10	10
source-10036-26	D1	10	0	0	6	6	6
source-10036-27	D2	10	0	0	16	16	16
source-10036-28	D2	10	0	0	9	9	9
source-10036-29	D2	10	2	0	13	13	13
source-10036-30	D2	10	2	0	13	13	13
source-10036-31	D2	10	2	0	12	12	12
source-10036-32	D2	10	2	0	13	11	14
source-10036-33	D3	10	4	0	20	20	20
source-10036-34	D3	10	4	0	22	22	22
source-10036-35	D3	10	4	0	22	18	134
source-10036-36	D3	10	4	0	22	22	22
source-10036-37	D2	10	0	0	14	12	60
source-10036-38	D4	10	0	0	27	17	786
source-10036-39	D3	10	0	0	20	14	264
source-10036-40	D1	10	0	0	5	5	7
source-10036-41	D4	10	2	0	33	15	4640
source-10036-42	D3	10	2	0	17	11	282
source-10036-43	D2	10	2	0	14	12	101
source-10036-44	D3	10	2	0	19	17	236
source-10036-45	D3	10	4	0	19	15	726
source-10036-46	D2	10	4	0	15	13	151
source-10036-47	D4	10	4	0	32	22	11933
source-10036-48	D2	10	4	0	14	10	311
source-10213-01	D1	2	0	0	2	2	2
source-10213-02	D1	2	0	0	1	1	1
source-10213-03	D1	2	0	0	1	1	1
source-10213-04	D1	2	2	0	6	6	6
source-10213-05	D1	2	2	0	4	4	4
source-10213-06	D1	2	2	0	5	5	5
source-10213-07	D1	2	2	0	5	5	5
source-10213-08	D1	2	0	0	2	2	2
source-10213-09	D1	2	0	0	1	1	1
source-10213-10	D1	2	0	0	1	1	1
source-10213-11	D1	2	2	0	4	4	4
source-10213-12	D1	2	2	0	2	2	2
source-10213-13	D1	2	2	0	3	3	3
source-10213-14	D1	2	2	0	3	3	3
source-10213-15	D1	2	0	0	1	1	1
source-10213-16	D1	2	0	0	2	2	2
source-10213-17	D1	2	0	0	1	1	1
source-10213-18	D1	2	2	0	4	4	4
source-10213-19	D1	2	2	0	5	5	5
source-10213-20	D1	2	2	0	4	4	4
source-10213-21	D1	2	2	0	3	3	3
source-10213-22	D1	2	0	0	1	1	1
source-10213-23	D1	2	0	0	1	1	1
source-10213-24	D1	2	0	0	2	2	2
source-10213-25	D1	2	2	0	3	3	3
source-10213-26	D1	2	2	0	4	4	4

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-10213-27	D1	2	2	0	5	5	5
source-10213-28	D1	2	2	0	4	4	4
source-omr-10242-01	D1	10	0	0	8	8	8
source-omr-10242-02	D2	10	0	0	14	14	14
source-omr-10242-03	D1	10	0	0	7	7	7
source-omr-10242-04	D2	10	0	0	11	11	11
source-omr-10242-05	D2	10	2	0	13	13	13
source-omr-10242-06	D2	10	2	0	16	16	16
source-omr-10242-07	D2	10	2	0	11	11	11
source-omr-10242-08	D2	10	2	0	15	15	15
source-omr-10242-09	D3	10	4	0	17	17	17
source-omr-10242-10	D2	10	4	0	16	16	16
source-omr-10242-11	D2	10	4	0	14	14	14
source-omr-10242-12	D3	10	4	0	17	17	17
source-omr-10242-13	D2	10	0	0	9	9	9
source-omr-10242-14	D1	10	0	0	7	7	7
source-omr-10242-15	D2	10	0	0	9	9	9
source-omr-10242-16	D1	10	0	0	8	8	8
source-omr-10242-17	D2	10	2	0	13	13	13
source-omr-10242-18	D2	10	2	0	16	16	16
source-omr-10242-19	D2	10	2	0	12	12	12
source-omr-10242-20	D2	10	2	0	12	12	12
source-omr-10242-21	D2	10	4	0	14	14	14
source-omr-10242-22	D2	10	4	0	13	13	13
source-omr-10242-23	D2	10	4	0	13	13	13
source-omr-10242-24	D2	10	4	0	15	15	15
source-omr-10242-25	D2	10	0	0	11	11	11
source-omr-10242-26	D1	10	0	0	7	7	7
source-omr-10242-27	D1	10	0	0	6	6	6
source-omr-10242-28	D2	10	0	0	10	10	10
source-omr-10242-29	D2	10	2	0	9	9	9
source-omr-10242-30	D3	10	2	0	17	17	17
source-omr-10242-31	D2	10	2	0	10	10	10
source-omr-10242-32	D2	10	2	0	12	12	12
source-omr-10242-33	D2	10	4	0	14	14	14
source-omr-10242-34	D2	10	4	0	14	14	14
source-omr-10242-35	D2	10	4	0	15	15	15
source-omr-10242-36	D2	10	4	0	14	14	14
source-omr-10242-37	D2	10	0	0	10	10	10
source-omr-10242-38	D1	10	0	0	8	8	8
source-omr-10242-39	D1	10	0	0	7	7	7
source-omr-10242-40	D1	10	0	0	8	8	8
source-omr-10242-41	D2	10	2	0	10	10	10
source-omr-10242-42	D2	10	2	0	11	11	11
source-omr-10242-43	D2	10	2	0	11	11	11
source-omr-10242-44	D2	10	2	0	12	12	12
source-omr-10242-45	D2	10	4	0	16	16	16
source-omr-10242-46	D3	10	4	0	17	17	17
source-omr-10242-47	D3	10	4	0	17	17	17
source-omr-10242-48	D3	10	4	0	18	18	18
source-omr-10265-01	D1	3	0	0	2	2	2
source-omr-10265-02	D1	3	0	0	1	1	1
source-omr-10265-03	D1	3	0	0	1	1	1
source-omr-10265-04	D1	3	0	0	3	3	3
source-omr-10265-05	D1	3	2	0	5	5	5
source-omr-10265-06	D1	3	2	0	7	7	7

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-omr-10265-07	D1	3	2	0	4	4	4
source-omr-10265-08	D1	3	2	0	5	5	5
source-omr-10265-09	D2	3	4	0	9	9	9
source-omr-10265-10	D2	3	4	0	10	10	10
source-omr-10265-11	D2	3	4	0	12	12	12
source-omr-10265-12	D2	3	4	0	9	9	9
source-omr-10265-13	D1	3	0	0	2	2	2
source-omr-10265-14	D1	3	0	0	2	2	2
source-omr-10265-15	D1	3	0	0	2	2	2
source-omr-10265-16	D1	3	0	0	1	1	1
source-omr-10265-17	D1	3	2	0	4	4	4
source-omr-10265-18	D1	3	2	0	7	7	7
source-omr-10265-19	D1	3	2	0	5	5	5
source-omr-10265-20	D1	3	2	0	5	5	5
source-omr-10265-21	D2	3	4	0	9	9	9
source-omr-10265-22	D2	3	4	0	9	9	9
source-omr-10265-23	D2	3	4	0	9	9	9
source-omr-10265-24	D2	3	4	0	9	9	9
source-omr-10265-25	D1	3	0	0	2	2	2
source-omr-10265-26	D1	3	0	0	3	3	3
source-omr-10265-27	D1	3	0	0	3	3	3
source-omr-10265-28	D1	3	0	0	2	2	2
source-omr-10265-29	D1	3	2	0	3	3	3
source-omr-10265-30	D1	3	2	0	4	4	4
source-omr-10265-31	D1	3	2	0	5	5	5
source-omr-10265-32	D1	3	2	0	5	5	5
source-omr-10265-33	D1	3	4	0	7	7	7
source-omr-10265-34	D1	3	4	0	7	7	7
source-omr-10265-35	D1	3	4	0	7	7	7
source-omr-10265-36	D1	3	4	0	8	8	8
source-omr-10265-37	D1	3	0	0	2	2	2
source-omr-10265-38	D1	3	0	0	3	3	3
source-omr-10265-39	D1	3	0	0	1	1	1
source-omr-10265-40	D1	3	0	0	1	1	1
source-omr-10265-41	D1	3	2	0	8	8	8
source-omr-10265-42	D1	3	2	0	6	6	6
source-omr-10265-43	D1	3	2	0	7	7	7
source-omr-10265-44	D1	3	2	0	6	6	6
source-omr-10265-45	D2	3	4	0	9	9	9
source-omr-10265-46	D2	3	4	0	9	9	9
source-omr-10265-47	D2	3	4	0	9	9	9
source-omr-10265-48	D1	3	4	0	8	8	8
source-omr-10269-01	D2	10	0	0	11	11	11
source-omr-10269-02	D2	10	0	0	13	13	13
source-omr-10269-03	D2	10	0	0	12	12	12
source-omr-10269-04	D2	10	0	0	9	9	9
source-omr-10269-05	D3	10	2	0	17	17	17
source-omr-10269-06	D3	10	2	0	18	18	18
source-omr-10269-07	D2	10	2	0	16	16	16
source-omr-10269-08	D2	10	2	0	15	15	15
source-omr-10269-09	D3	10	4	0	22	22	22
source-omr-10269-10	D3	10	4	0	21	21	21
source-omr-10269-11	D3	10	4	0	19	19	19
source-omr-10269-12	D3	10	4	0	20	20	20
source-omr-10269-13	D1	10	0	0	6	6	6
source-omr-10269-14	D2	10	0	0	11	11	11

Episode	Tier	Rack	Jobs	Done	C^*	h	A* exp.
source-omr-10269-15	D2	10	0	0	10	10	10
source-omr-10269-16	D1	10	0	0	7	7	7
source-omr-10269-17	D2	10	2	0	14	14	14
source-omr-10269-18	D2	10	2	0	15	15	15
source-omr-10269-19	D2	10	2	0	11	11	11
source-omr-10269-20	D2	10	2	0	12	12	12
source-omr-10269-21	D3	10	4	0	18	18	18
source-omr-10269-22	D2	10	4	0	16	16	16
source-omr-10269-23	D2	10	4	0	13	13	13
source-omr-10269-24	D2	10	4	0	14	14	14
source-omr-10269-25	D2	10	0	0	14	14	14
source-omr-10269-26	D1	10	0	0	7	7	7
source-omr-10269-27	D2	10	0	0	11	11	11
source-omr-10269-28	D2	10	0	0	10	10	10
source-omr-10269-29	D2	10	2	0	12	12	12
source-omr-10269-30	D2	10	2	0	13	13	13
source-omr-10269-31	D2	10	2	0	14	14	14
source-omr-10269-32	D2	10	2	0	12	12	12
source-omr-10269-33	D2	10	4	0	15	15	15
source-omr-10269-34	D3	10	4	0	17	17	17
source-omr-10269-35	D2	10	4	0	14	14	14
source-omr-10269-36	D2	10	4	0	15	15	15
source-omr-10269-37	D2	10	0	0	12	12	12
source-omr-10269-38	D2	10	0	0	13	13	13
source-omr-10269-39	D2	10	0	0	12	12	12
source-omr-10269-40	D1	10	0	0	8	8	8
source-omr-10269-41	D2	10	2	0	15	15	15
source-omr-10269-42	D2	10	2	0	13	13	13
source-omr-10269-43	D2	10	2	0	12	12	14
source-omr-10269-44	D2	10	2	0	16	16	16
source-omr-10269-45	D2	10	4	0	16	16	16
source-omr-10269-46	D3	10	4	0	18	18	18
source-omr-10269-47	D3	10	4	0	17	17	17
source-omr-10269-48	D2	10	4	0	15	15	15
source-10014-context-0-01	D3	10	1	0	18	18	23
source-10014-context-1-01	D3	10	1	0	18	18	18
source-10014-context-0-02	D3	10	1	0	22	22	31
source-10014-context-1-02	D3	10	1	0	22	22	22
source-10036-context-0-01	D2	10	0	0	15	15	15
source-10036-context-1-01	D2	10	0	0	15	15	15
source-10036-context-0-02	D2	10	0	0	16	16	16
source-10036-context-1-02	D2	10	0	0	16	16	16
source-10036-intervention-2-context-0-01	D3	10	0	0	17	17	17
source-10036-intervention-2-context-1-01	D3	10	0	0	17	17	17
source-10036-intervention-2-context-0-02	D2	10	0	0	15	15	15
source-10036-intervention-2-context-1-02	D2	10	0	0	15	15	15